Minimum induced drag of superelliptical closed lifting systems with a loaded central body

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Abstract

Closed ("box") wings have been known since Prandtl to reduce induced drag relative to planar wings of equal span, but practical ring-wing aircraft concepts raise two questions the classical literature does not answer together: how do smoothly curved closed traces compare with the rectangular box at equal height-to-span ratio, and what does it cost to make a central cabin body carry lift inside the ring? Using a Trefftz-plane minimum-induced-drag formulation, discretised with piecewise-constant circulation panels and solved as an equality-constrained quadratic program, this study (i) validates against three classical results (planar wing $e = 1$, Prandtl's box wing $e \approx 1.46$ at $h/b = 0.2$, circular ring $e = 2$); (ii) shows that the optimal span efficiency of an elliptical ring follows $e = 1 + h/b$ across the full range, with Richardson extrapolation of the observed first-order grid convergence reducing the residual to 10^{-6}—strong evidence of an exact law; (iii) maps the superellipse family $|y/a|^p + |z/c|^p = 1$ between ellipse and box, finding that a $p = 4$ "squircle" captures roughly three quarters of the box-wing benefit and $p = 6$ captures 85%; and (iv) quantifies the penalty of loading a central body, which is smallest when the body hugs either arc of the ring and largest at mid-height. A finite-chord vortex-lattice cross-check shows the ideal efficiencies are practically attainable: untwisted rings at uniform incidence already achieve 98.6–100% of the Trefftz optimum, and less than one degree of twist recovers the remainder. A design point combining these findings—a $p = 4$ superelliptical ring of $h/b = 0.5$ with a low-set cabin carrying 15% of total lift—achieves $e = 1.61$, a 38% induced-drag reduction relative to an ideal planar wing.

1 Introduction

Induced drag—the drag due to lift—accounts for roughly a third of the cruise drag of a typical transport aircraft and close to half of its drag at climb. Munk [1] showed that for a planar wing the minimum induced drag is achieved by an elliptical lift distribution, giving the familiar

$$D_i \;=\; \frac{L^2}{\pi q b^2 e}, \tag{1}$$

with span efficiency $e = 1$. Nonplanar and closed lifting systems can exceed $e = 1$: Prandtl [2] identified the rectangular "box wing" as the *best wing system*, with an efficiency that grows with the height-to-span ratio h/b (approximately $e \approx 1.46$ at $h/b = 0.2$), and Cone [3] systematically studied nonplanar systems, including annular wings. The subject remains active in the joined-wing and PrandtlPlane literature [4, 5].

Ring-wing *aircraft* concepts, however, differ from the idealised closed traces studied classically in two ways. First, for structural, manufacturing, and aesthetic reasons a practical ring is smoothly curved rather than rectangular; the efficiency of intermediate shapes between the ellipse and the box does not appear to have been mapped. Second, such concepts typically place the cabin as a lifting body *inside* the ring, joined to it fore and aft; in the Trefftz plane this appears as a detached central lifting element, and the induced-drag cost of requiring it to carry a share of the total lift has, to the author's knowledge, not been quantified.

This paper addresses both questions with a single validated numerical tool. The contributions are:

1. a numerically supported conjecture that the optimal span efficiency of an elliptical closed trace is exactly $e = 1 + h/b$;
2. the efficiency map of the superellipse family $|y/a|^p + |z/c|^p = 1$ between ellipse ($p = 2$) and box ($p \to \infty$), showing rapid saturation of the benefit with p;
3. the induced-drag penalty of a loaded central body as a function of its lift share and of its vertical position, yielding a simple design rule: blend the body into an arc of the ring, do not suspend it at mid-height;
4. a finite-chord vortex-lattice cross-check showing that these ideal efficiencies are attainable by real surfaces: untwisted rings reach 98.6–100% of the optimum and sub-degree twist closes the gap;
5. a combined design point demonstrating $e = 1.61$ with realistic cabin loading.

2 Method

2.1 Trefftz-plane formulation

Far downstream of a steady lifting system the wake tends to a two-dimensional vortex sheet whose trace C in the transverse (Trefftz) plane determines the induced drag entirely [1]. For a circulation distribution $\Gamma(s)$ along the trace (arc length s), lift and induced drag are

$$L = \rho V \int_C \Gamma \,\mathrm{d}y, \qquad D_i = \frac{\rho}{2} \int_C \Gamma \, w_n \,\mathrm{d}s, \tag{2}$$

where w_n is the sheet-normal component of the velocity induced in the Trefftz plane by the trailing vorticity $-\mathrm{d}\Gamma/\mathrm{d}s$. Minimising D_i at fixed L defines the optimal loading and, through Eq. (1) with b the overall span of the system, the span efficiency e reported throughout. Because e is a ratio of the same quadratic functionals for every geometry, it is independent of the choices $\rho = V = 1$ made here.

2.2 Discretisation

The trace is divided into n panels carrying piecewise-constant circulations Γ_i. Each panel sheds two-dimensional point vortices of strength $-\Gamma_i$ and $+\Gamma_i$ at its endpoints; at an edge shared by two panels these superpose to the physical shed strength $\Gamma_i - \Gamma_{i+1}$, while free ends of open components automatically carry the full tip vortex. Control points lie at panel midpoints. Substituting into Eq. (2) gives

$$D_i = \mathbf{\Gamma}^{\mathsf{T}} A \, \mathbf{\Gamma}, \qquad L = \mathbf{b}^{\mathsf{T}} \mathbf{\Gamma}, \tag{3}$$

with A symmetrised and $b_i = \Delta y_i$. Lift-equality constraints (total lift, and optionally the lift of an individual component) are collected in $B^{\mathsf{T}}\boldsymbol{\Gamma} = \mathbf{d}$ and the optimum found from the KKT system

$$\begin{pmatrix} 2A & -B \\ B^{\mathsf{T}} & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{\Gamma} \\ \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{d} \end{pmatrix}. \tag{4}$$

Closed polygons are resampled to uniform arc length (equal panel sizes avoid corner ill-conditioning of the point-vortex kernel); open traces use cosine spacing. A Tikhonov term of 10^{-12} on the diagonal of A guards conditioning. Resolutions up to $n = 1536$ are used; convergence is reported in Sec. 3.

Geometries studied: the planar wing; the rectangular box wing of height h; the elliptical ring ($y = a\cos\theta$, $z = \frac{h}{2}(1 + \sin\theta)$); the superellipse family $|y/a|^p + |z/c|^p = 1$ with $c = h/2$; and rings augmented by a detached central segment (the Trefftz trace of a cabin body joined to the ring fore and aft) of width 0.4–0.5 b at height $z = z_f h$.

3 Validation

Three classical results serve as validation targets (Table 1). The planar wing recovers Munk's $e = 1$ to 0.3% at $n = 400$. The box wing at $h/b = 0.2$ converges to $e = 1.472$, consistent with Prandtl's approximate value ≈ 1.46 and with later exact treatments [2, 5]. The circular ring converges to the classical $e = 2$ [3] to four decimal places, a stringent test since it involves the full closed-trace kernel. Consistency checks: the superellipse geometry at $p = 2$ reproduces the independent ellipse parameterisation to 1.5×10^{-3}, and an unconstrained central body takes up zero loading, leaving e unchanged, as Munk's stagger theorem requires.

The discretisation error of the kernel is cleanly first order: the planar-wing bias $e - 1$ falls from $+6.3 \times 10^{-3}$ at $n = 200$ to $+3.9 \times 10^{-4}$ at $n = 3200$, halving at each doubling of resolution (observed ratio 2.00). Absolute efficiencies reported at production resolutions therefore carry a systematic bias below +0.3%, while efficiency *ratios* between geometries computed on matched grids are unaffected. Corner resolution is likewise benign even for the squarest rings studied: at $p = 12$, $h/b = 0.5$, the efficiency changes by only 1.6×10^{-4} between $n = 384$ and $n = 3072$.

Table 1: Validation against classical results.

Case	Computed	Classical	Source
Planar wing ($n = 400$)	1.0031	1.0000	Munk [1]
Box wing, $h/b = 0.2$ ($n_{\text{span}} = 64 \ldots 512$)	$1.4737 \rightarrow 1.4721$	$\approx$1.46	Prandtl [2]
Circular ring ($n = 128 \ldots 1024$)	$1.9994 \rightarrow 2.0000$	2.0	Cone [3]

4 Results

4.1 Ellipse versus box, and the $e = 1 + h/b$ law

Figure 1 shows the optimal span efficiency of the box wing and the elliptical ring for h/b from 0.05 to 1. The box wing is superior at every height, consistent with its status as the best wing system: at $h/b = 0.3$ it reaches $e = 1.658$ against the ellipse's 1.302.

The elliptical results, however, collapse onto a strikingly simple relation. At $n = 1536$ the computed efficiency satisfies

$$e_{\text{ellipse}} = 1 + \frac{h}{b} \tag{5}$$

with a maximum deviation of 7.9×10^{-4} over $h/b \in [0.1, 1]$. The deviation decreases monotonically with h/b and vanishes at the circle (where Eq. (5) gives the exact classical value $e = 2$). A grid-convergence study at $h/b = 0.3$ shows the deviation halving at each doubling of resolution ($+2.9 \times 10^{-3}$ at $n = 384$ down to $+3.7 \times 10^{-4}$ at $n = 3072$); Richardson extrapolation of the observed first-order sequence drives it to $+1 \times 10^{-6}$, i.e. the relation holds to the numerical precision available here. The author has not located Eq. (5) in the accessible literature for elliptical closed traces; closed-form treatments of *circular* annular wings exist [3, 7], and the relation may be derivable from, or implicit in, such analyses—it is offered as a numerically supported conjecture, plausibly provable in elliptic coordinates, with the literature question flagged for the reader.

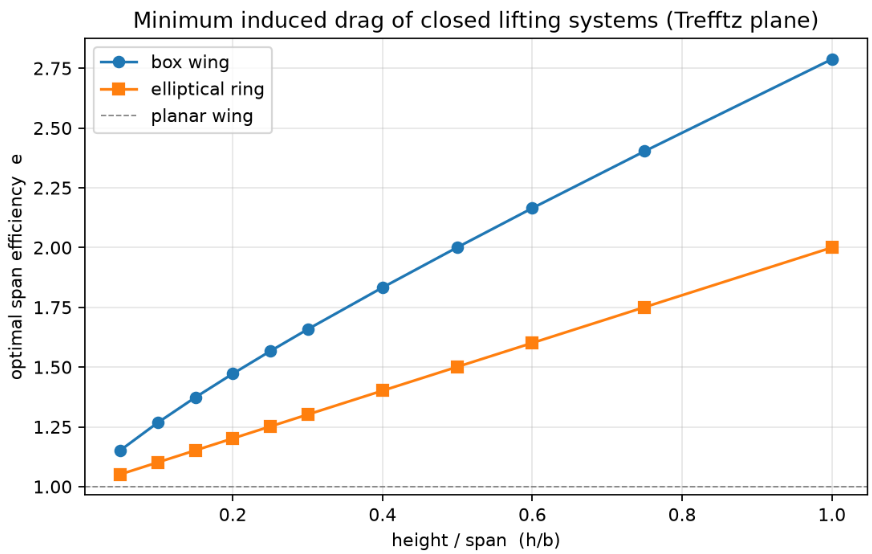

Figure 1: Optimal span efficiency of the box wing and the elliptical ring versus height-to-span ratio. The elliptical ring follows $e = 1 + h/b$ to within 8×10^{-4}.

4.2 The superellipse family: squaring the ring

Figure 2 maps $e(p)$ for the superellipse family at $h/b = 0.3$ and 0.5. Defining the *benefit capture* $(e - 1)/(e_{\text{box}} - 1)$, the gain saturates rapidly with p: $p = 3$ captures about two thirds of the box benefit, $p = 4$ about three quarters (73% at $h/b = 0.3$, 76% at $h/b = 0.5$), $p = 6$ some 83–85%, and $p = 12$ over 92%. The practical implication is that the sharp corners of a true box wing—structurally heavy, prone to interference drag in viscous flow, and difficult to manufacture—are unnecessary: a gently squared ring of $p = 4$–6 obtains most of the ideal-flow advantage while remaining smooth everywhere.

4.3 Cost of loading the central body

Figure 3 quantifies the penalty of forcing a central body (width $0.5\,b$, mid-height) to carry a fraction f of the total lift. The optimum assigns the body zero load; the penalty grows quadratically in f

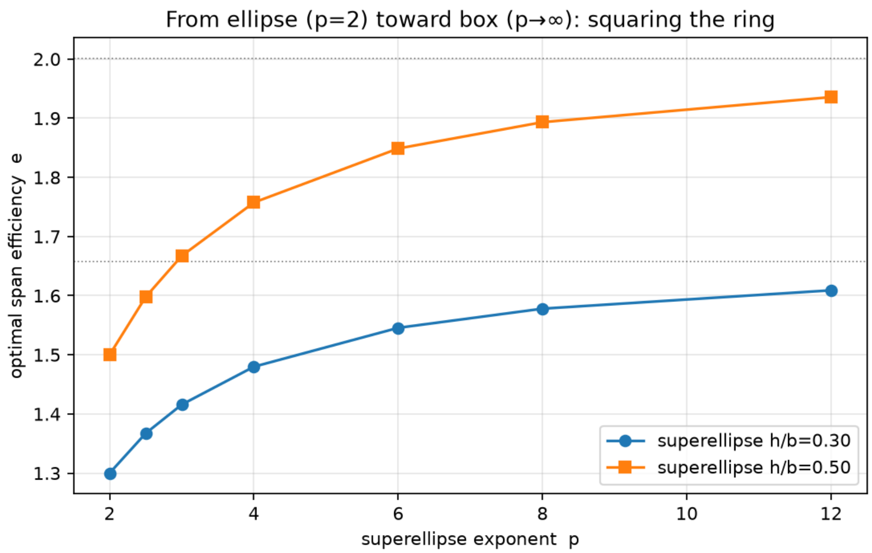

Figure 2: Span efficiency versus superellipse exponent p at two height-to-span ratios. Dotted lines mark the corresponding box-wing values ($p \to \infty$).

and is severe beyond $f \approx 0.15$: at $h/b = 0.3$, efficiency falls from 1.302 ($f = 0$) to 1.143 at $f = 0.2$ and drops below the planar-wing value at $f \approx 0.3$. A closed wing can therefore be squandered entirely by asking its cabin to lift like a wing.

The vertical placement study (Fig. 4) explains why and softens the verdict. With $f = 0.15$ fixed, the penalty is largest at mid-height ($e = 1.145$) and smallest when the body hugs either arc ($e = 1.226$ at $z/h = 0.1$), the curve being symmetric. Near an arc the body's shed vorticity partially merges with the ring's own optimal sheet; at mid-gap it is maximally exposed. The design rule is simple: *blend the cabin into an arc of the ring; never suspend it at mid-height.*

Both conclusions are robust to the assumed body width, and the width itself turns out to matter strongly. Repeating the placement study with body widths of $0.3\,b$, $0.5\,b$, and $0.7\,b$ preserves the monotone arc-hugging ordering in every case; but at fixed placement and lift share the penalty falls sharply as the body widens (at mid-height with $f = 0.2$: $e = 0.88$ at width $0.3\,b$ versus 1.23 at $0.7\,b$), because a narrow body must shed its vorticity in a concentrated, expensive pair. This adds a second design rule: *if the central body must carry lift, make it wide*—a natural fit for lifting-body cabins.

4.4 Finite-chord cross-check: are the ideal efficiencies attainable?

Trefftz-plane results are bounds; whether a finite-chord surface can realise them is a separate question. To answer it, a classical horseshoe-vortex lattice method [6] was built on the same trace geometries: strips extruded in the streamwise direction (144 around the loop, 8 chordwise panels, chord $c/b = 0.1$), boundary condition applied at the three-quarter-chord collocation points, and twist introduced through the right-hand side so the strip loading is exactly linear in the twist schedule. Crucially, the strip circulations are evaluated with the *same* Trefftz functionals used for the ideal results, so achieved and ideal efficiencies are measured by an identical ruler. The method passes three gates: an elliptical-planform wing of $AR = 8$ recovers 99.9% of its same-grid ideal with lift-curve slope 4.80 per radian (lifting-surface level, slightly below the Helmbold value 5.03); near-field Kutta–Joukowski lift agrees with the Trefftz lift to 0.4%; and the results are independent

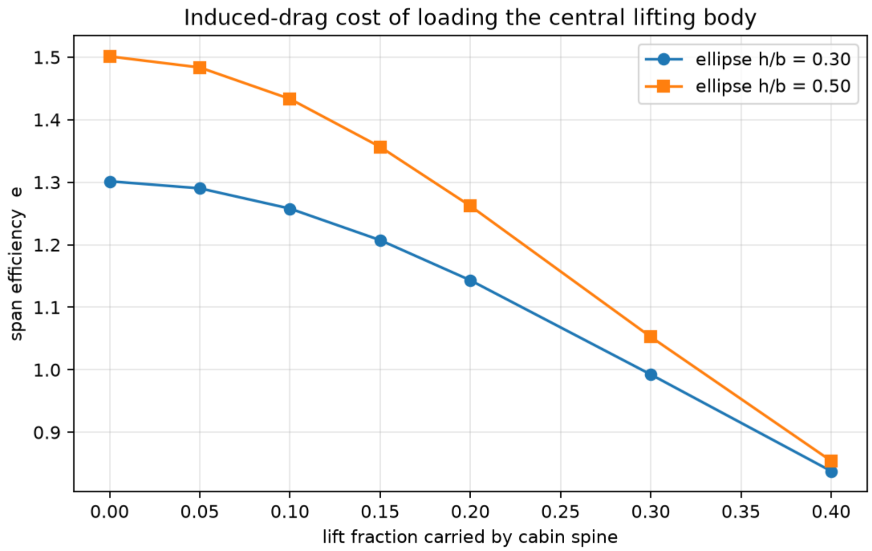

Figure 3: Efficiency versus lift fraction carried by a mid-height central body, for two ring heights.

of chordwise resolution to 2×10^{-4} between 4 and 12 panels.

Table 2: Finite-chord VLM versus Trefftz ideal (144 strips, $c/b = 0.1$, $\alpha = 5°$). Percentages are fractions of the same-grid ideal.

Ring	h/b	Ideal e	Untwisted e	Optimised e	Max twist
Circle	1.0	1.9995	1.9995 (100.0%)	1.9995 (100.0%)	0.00°
Ellipse	0.5	1.5061	1.4851 (98.6%)	1.5061 (100.0%)	0.76°
Superellipse p=4	0.5	1.7569	1.7413 (99.1%)	1.7569 (100.0%)	0.63°

Table 2 contains the central result of this section: a ring wing at *uniform, untwisted* incidence already achieves 98.6–100% of its ideal efficiency, and a twist schedule of less than one degree of amplitude recovers the remainder (Fig. 5). The near-optimality of the untwisted ring has a clean explanation. The flow-tangency condition at uniform incidence α forces a surface normalwash proportional to n_z, the vertical component of the local surface normal—which is precisely the form of Munk's minimum-drag criterion for the Trefftz-plane normalwash. For the circular ring the symmetry is exact and the untwisted loading *is* the optimal loading to numerical precision; for less symmetric traces the near-field/far-field distinction leaves a residual of one to two percent. The conclusion is unusually favourable: unlike the planar wing, whose rectangular planform loses 3% and needs significant washout to recover it, a closed ring is nearly self-optimising. Sensitivity to chord is likewise mild: the untwisted superellipse ring stays within 97.9–99.7% of ideal for c/b between 0.05 and 0.20. The result also survives stagger, which practical ring configurations use for trim and cabin integration: offsetting the upper arc aft and the lower arc forward by up to $0.5\,b$ in total reduces the untwisted recovery only from 99.1% to 98.1%, and the optimised twist remains small (2.1°)—consistent with Munk's stagger theorem, which guarantees the ideal bound itself is stagger-independent. (The twist-optimised entries in Table 2 match the ideal by construction—with one twist variable per strip the target loading is matched essentially exactly; the substantive finding is that the twist required is smaller than typical manufacturing tolerances on jig twist.)

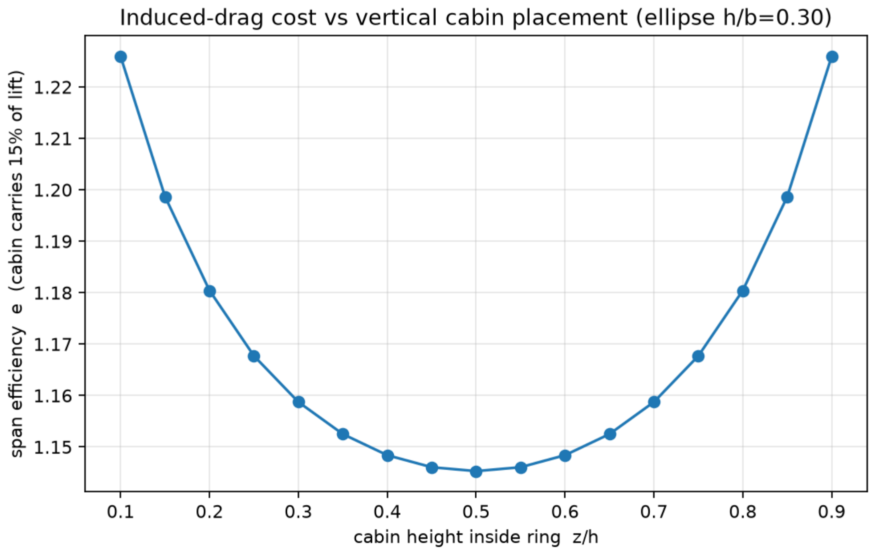

Figure 4: Efficiency versus vertical position of the central body at a fixed 15% lift share (ellipse, $h/b = 0.3$). Mid-height is the worst placement.

4.5 A combined design point

Applying all three findings to a hypothetical transport configuration—a superelliptical ring with $p = 4$, $h/b = 0.5$, and a cabin of width $0.5\,b$ blended low in the ring at $z/h = 0.15$ carrying 15% of the lift—gives $e = 1.61$, against 1.76 for the same ring unloaded, 1.50 for the plain ellipse, and 2.00 for the ideal box. Relative to an ideal planar wing of equal span, the design point represents a 38% reduction in induced drag with a realistic cabin lift share included. Figure 6 shows the optimal circulation distribution at this design point.

5 Discussion and limitations

The analysis is inviscid, linear, and assumes the optimal (fully relaxed) load distribution on a rigid, non-contracting wake; it therefore bounds what any real configuration can achieve rather than predicting what a particular one will. Viscous interference at wing–body junctions, profile drag of the added wetted area of a closed system, structural weight, stability and trim constraints, and compressibility are all outside its scope. These caveats cut both ways: the box wing's corner regions are precisely where viscous interference is worst, so the superellipse's advantage over the box is likely understated in real flow, strengthening the case for smooth closed traces. The $e = 1 + h/b$ relation, if exact, would give designers a closed-form anchor for curved closed systems analogous to Prandtl's box-wing charts.

Two further limitations deserve emphasis. First, while Sec. 4.4 shows that the ideal loadings are attainable by finite-chord rings with sub-degree twist, that demonstration is itself inviscid and untrimmed; pitch trim and stability constraints will perturb the achievable loading, though the margin documented there (98.6–100% before any twist at all) leaves room for such perturbations. Second, the central-body results model the cabin trace as a flat segment; a cambered lifting body of finite thickness will differ in detail, though the geometric trend—arc-hugging beats mid-gap—follows from the structure of the induced-velocity kernel and should be robust.

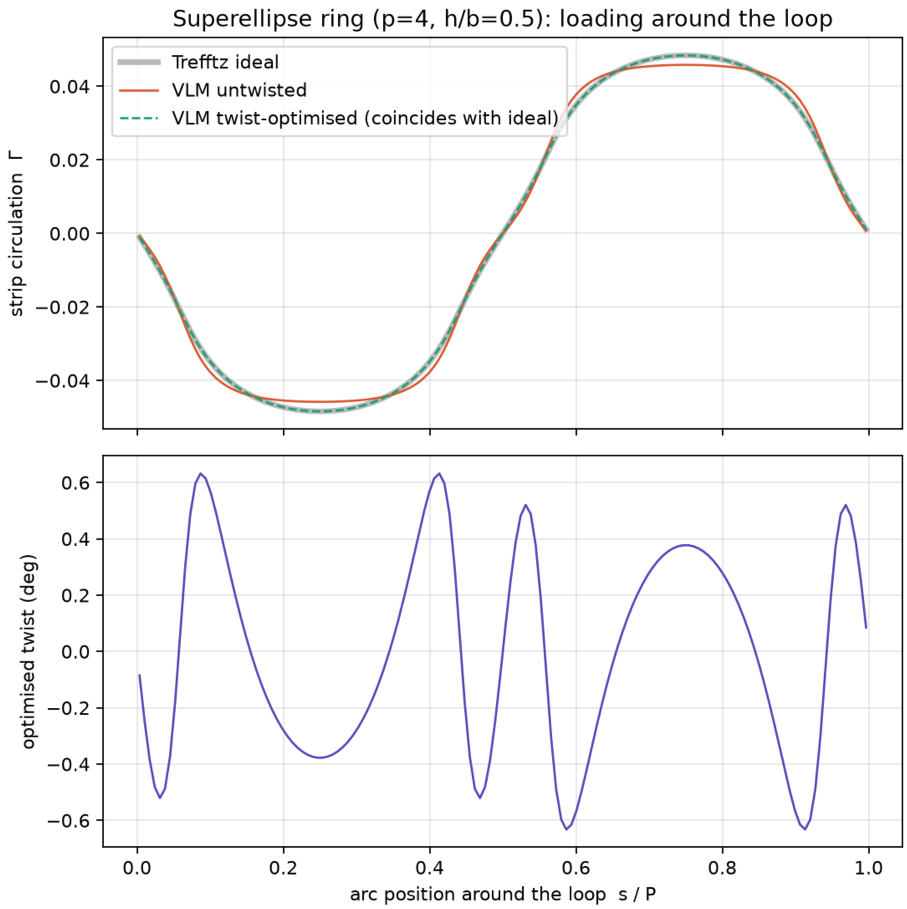

Figure 5: Superellipse ring ($p = 4$, $h/b = 0.5$): strip circulation around the loop for the Trefftz ideal, the untwisted VLM, and the twist-optimised VLM (top), and the sub-degree optimised twist schedule (bottom).

6 Conclusion

A validated Trefftz-plane optimiser was used to map the induced-drag performance of closed lifting systems between the two classical extremes. The elliptical ring obeys $e = 1+h/b$ to within 8×10^{-4}, proposed here as an exact law; the superellipse family shows that a $p = 4$–6 "squircle" ring captures 73–85% of the box wing's benefit without its corners; and loading a central cabin body is cheap only near the ring's arcs and only up to a lift share of about 10–15%. A finite-chord vortex-lattice cross-check, sharing the same Trefftz functionals, shows these bounds are practically attainable: untwisted rings at uniform incidence reach 98.6–100% of the ideal (the circle exactly, by symmetry, since uniform incidence imposes Munk's optimality criterion in near-field form), and less than one degree of twist closes the gap. Together these results reduce ring-wing aircraft design from aesthetic speculation to three quantitative rules, and the combined design point ($e = 1.61$ with cabin loading) shows the concept survives its own constraints. Future work: an analytical proof of Eq. (5); trim and static-stability constraints in the vortex-lattice model; and a subscale flight demonstrator to measure achieved (as opposed to ideal) efficiency.

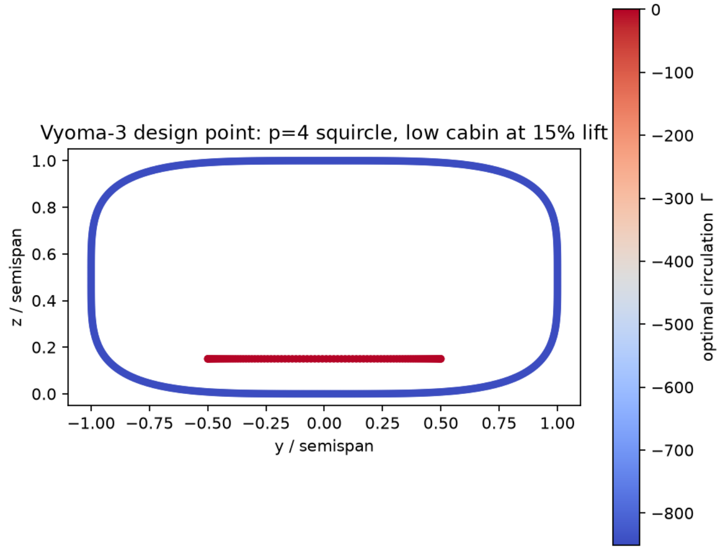

Figure 6: Optimal circulation at the combined design point: $p = 4$ superellipse, $h/b = 0.5$, low-set central body at 15% lift share.

Data and code availability

The Trefftz-plane solver and the vortex-lattice cross-check (∼500 lines of Python in total), all run scripts, and all data tables are available at [repository URL / Zenodo DOI].

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